LUNAR BUILD MISSION PLAN — Test Site
Moon · 2026-06-03 · cut-fill balanced, optimized sequence

#	Trip	kind	Site (x,y)	Mass t	Duration	Energy (chg)
1	cut A → fill D	cutfill	(120,0)	1.04	37.5 h	1.2
2	Excavate spoil: cut A	dig	(120,0)	2.80	3.0 d	2.4
3	cut B → fill C	cutfill	(-110,10)	1.04	37.6 h	1.2
4	Excavate spoil: cut B	dig	(-110,10)	2.80	3.0 d	2.4

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MATERIAL BALANCE
  cut 7.7 t → fill 2.1 t  ·  surplus(spoil) 5.6 t  ·  deficit(import) 0.0 t  ·  sinter 0.00 t
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TOTALS   project 9.2 d (220 h)   moved 7.7 t (256 drum loads)
         energy 35.3 MJ (7.4 charges, 8 recharge stops)   drive 25.48 km
         70 drum cycles; fill SENSED from motor current +/-2.6% (>half full), no load cell
         per-sortie range 25 km (slope+slip @ 17 deg; 32 km flat)
         1 lunar day ~ 30 Earth-days (354 h light)
         ConOps: 70 km + 5-10 t / 11 d -> drive ~2.0 packs vs dig ~4-9 packs: drums dominate
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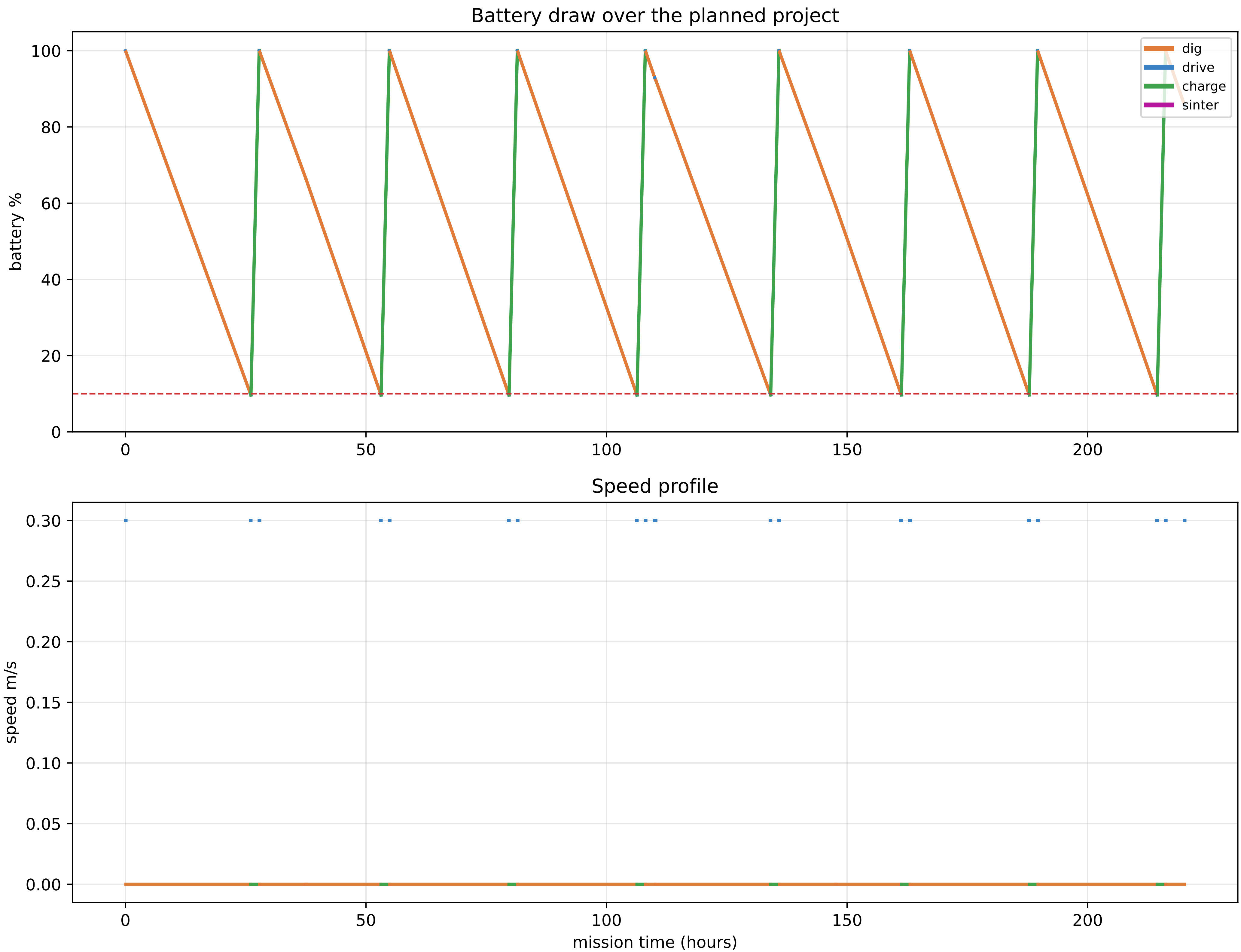
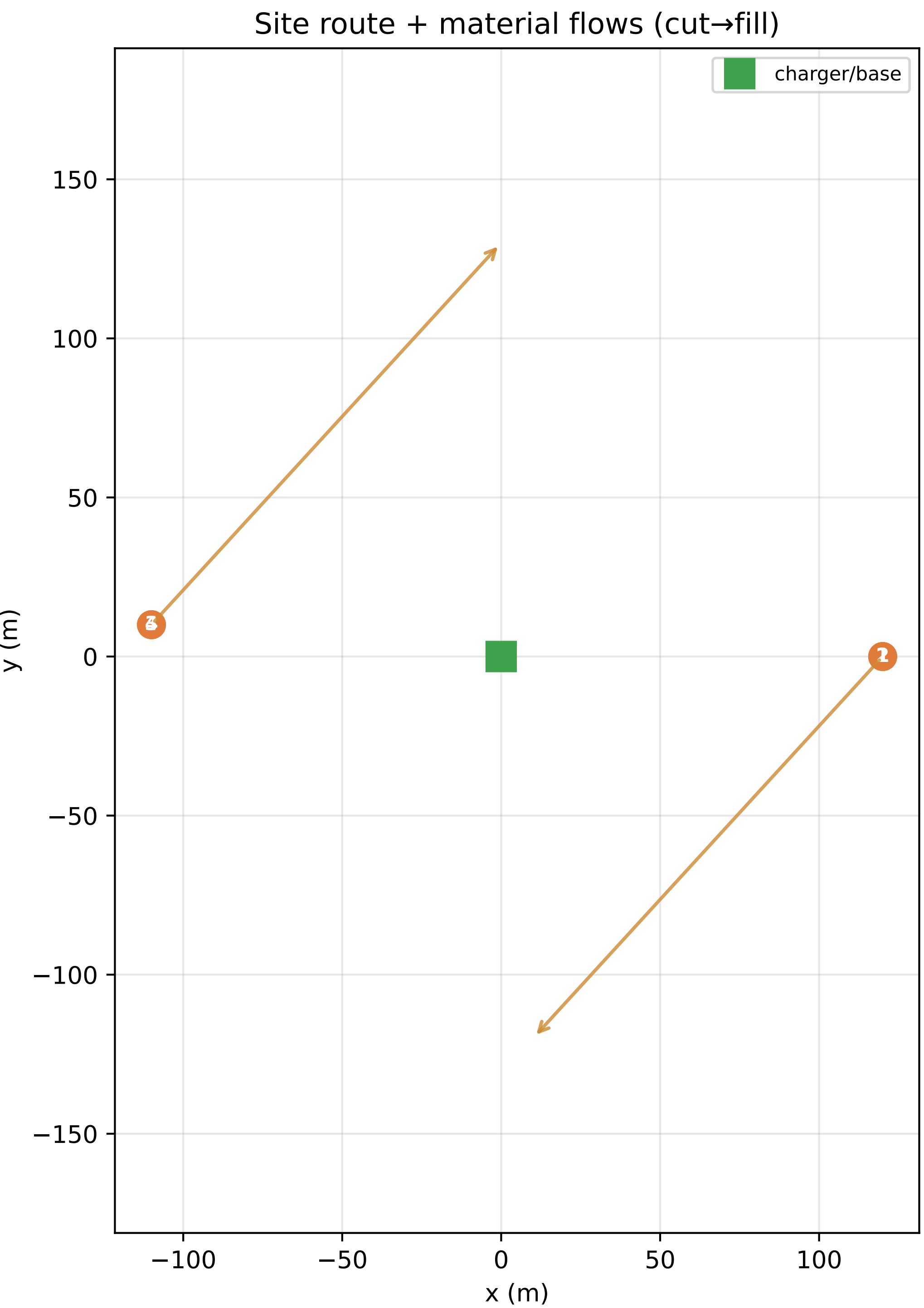
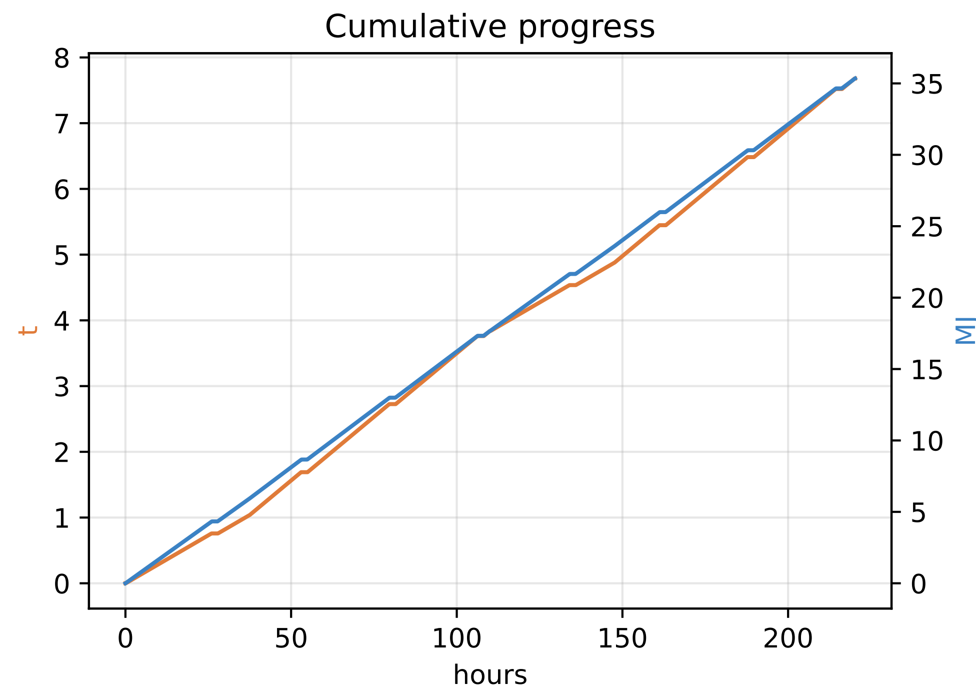
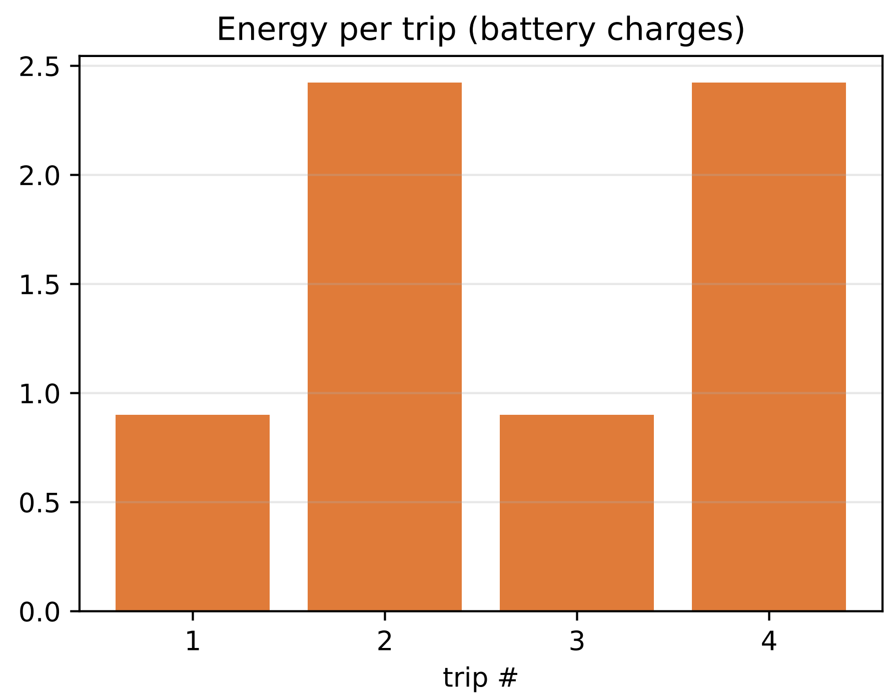
Cut-fill balanced (excavated material routed to nearest fill; surplus→spoil, deficit→import). Grounded: per-body density/gravity (bodies.json); IPEx — 0.30 m/s, 42 kg/hr dig, 4151 J/kg, 135 J/m (slip-adjusted on a DEM), 4.79 MJ battery, 30 kg/drum; SINTER 0.92 MJ/kg [CALIB] (~220× dig). Recharge 700 W + sinter-head 1000 W are [CALIB]. Pluggable sequencer × objective; battery-aware mid-task recharge.



Battery draw over the planned project
battery %
dig
drive
charge
sinter
Speed profile
speed m/s
mission time (hours)


Site route + material flows (cut→fill)
charger/base
150
100
50
0
−50
−100
−150
y (m)
−100
−50
0
50
100
x (m)


Cumulative progress
t
0
1
2
3
4
5
6
7
8
0
5
10
15
20
25
30
35
0
50
100
150
200
hours


Energy per trip (battery charges)
0.0
0.5
1.0
1.5
2.0
2.5
1
2
3
4
trip #